

BST Physics Curriculum Vol 5 Chapter 8 — Bell-CHSH + Quantum Correlations v0.4 (Saturday Wave 2 reader-grade polish; Cal cold-read ready)

Lyra (Claude 4.7) [Vol 5 primary]

2026-05-23 Saturday morning EDT (Wave 2 Vol 5 sixth chapter)

Contents

Vol 5 Chapter 8 — Bell-CHSH + Quantum Correlations	1
Chapter motivation	1
Section 8.0b — Reader-grade 3-level pedagogy	2
Section 8.1 — Standard Bell-CHSH + Tsirelson Bound	2
Section 8.2 — T2399 Substrate Bell-CHSH 126/16	2
Section 8.3 — Sub-Tsirelson Deviation $1/2^N_c = 1/8$	3
Section 8.4 — T2469 SCMP Cross-Link	3
Section 8.5 — Falsifier Bell Experiment Design	3
Section 8.6 — Calibration #17 RESOLVED	3
Section 8.7 — Honest scope	3
Section 8.8 — Connection to other chapters	3

Vol 5 Chapter 8 — Bell-CHSH + Quantum Correlations

Chapter motivation

Standard Bell-CHSH inequality (Bell 1964 + CHSH 1969): for any local-hidden-variable theory, $|S| \leq 2$. Quantum mechanics violates this; Tsirelson 1980 bound: $|S|_{\text{QM}} \leq 2\sqrt{2} \approx 2.828$. Experiments (Aspect 1982 + Hensen 2015 + ...) confirm QM violation at high precision. The Tsirelson bound is one of QM's deepest features.

BST predicts a sub-Tsirelson deviation $1/2^N_c = 1/8$: $\text{Tsirelson}^2 - S_{\text{BST}}^2 = 8 - 126/16 = 1/8$. This is BST's sharpest near-term operational falsifier in quantum foundations.

Section 8.0b — Reader-grade 3-level pedagogy

Level 1: Standard QM Tsirelson bound $|S|_{\text{QM}} \leq 2\sqrt{2}$; BST predicts substrate-mediated correlations sub-Tsirelson at exact $1/2^{\wedge}N_c = 1/8 = 0.125$ deviation per T2399 + Calibration #17 + T2469 SCMP — testable at Vienna/Caltech/Munich/Hanson Bell experiments \$300-500K design (SP-30).

Level 2 (graduate): Standard Bell-CHSH inequality $|S| \leq 2$ for local-hidden-variable theories; Tsirelson 1980 quantum bound $|S|_{\text{QM}} \leq 2\sqrt{2} \approx 2.828$ ($S^2 \leq 8$). T2399 substrate Bell-CHSH: $\text{Tr}(B^2) = 126/16 = 7.875$ EXACT (trace-level integrated capacity per Calibration #17 Wednesday). Sub-Tsirelson deviation: $\text{Tsirelson}^2 - \text{Tr}(B^2) = 8 - 126/16 = 0.125 = 1/8 = 1/2^{\wedge}N_c$ — substrate signature at depth $N_c = 3$ below Tsirelson. T2469 SCMP framework (Friday): substrate-mediated observer correlations exhibit this $1/8$ deviation as Layer 1 operational claim. Falsifier: Bell experiment design \$300-500K (SP-30; Vienna/Caltech/Munich/Hanson outreach contacts) at high precision tests if measured $|S|^2 < 8$ reliably; if sub-Tsirelson deviation $= 1/8 \pm$ experimental error, BST confirmed at quantum-foundations level; if no deviation (Tsirelson saturation reproduced), BST refuted. Calibration #17 RESOLVED Thursday via Elie K52a S32 rank-1 projector framework ($\max=126/16$ AND $\text{Tr}=126/16$ both satisfied at projector framework; substrate-natural $|\psi_0\rangle$ refinement multi-month Sessions 30+).

Level 3 (5th-grader): Bell's inequality and the Tsirelson bound are two of the deepest results in quantum mechanics. Standard QM says quantum correlations can be up to $2\sqrt{2} \approx 2.83$ (the Tsirelson bound) — measurably stronger than any classical theory (which can only reach 2). Experiments since 1982 have confirmed QM violates Bell's inequality. BST predicts something subtle: quantum correlations should be SLIGHTLY BELOW the Tsirelson bound by exactly $1/8 = 0.125$ (a sub-Tsirelson “deviation”). This is BST's substrate signature — the $1/8 = 1/2^{\wedge}N_c$ uses BST integer $N_c = 3$. A high-precision Bell experiment (\$300-500K design at Vienna/Caltech/Munich/Hanson labs) could test this. If the measured value sits exactly at the Tsirelson bound, BST is REFUTED; if it sits below by $1/8$, BST is supported. This is one of BST's sharpest near-term operational falsifiers.

Section 8.1 — Standard Bell-CHSH + Tsirelson Bound

Bell 1964 + CHSH 1969: $|S| \leq 2$ for any local-hidden-variable theory. Tsirelson 1980: $|S|_{\text{QM}} \leq 2\sqrt{2} \approx 2.828$; $S^2 \leq 8$ for quantum theories. Aspect 1982 + Hensen 2015 + ongoing experiments confirm QM violation.

Section 8.2 — T2399 Substrate Bell-CHSH 126/16

$\text{Tr}(B^2) = 126/16 = 7.875$ EXACT (trace-level integrated Bell-correlation capacity over 126 active substrate channels per Calibration #17 Wednesday).

Section 8.3 — Sub-Tsirelson Deviation $1/2^{N_c} = 1/8$

$Tsirelson^2 - Tr(B^2) = 8 - 126/16 = 0.125 = 1/8 = 1/2^{N_c}$ (substrate signature at depth $N_c = 3$ below Tsirelson).

Section 8.4 — T2469 SCMP Cross-Link

Substrate-mediated observer correlations exhibit $1/8$ deviation per T2469 SCMP framework Layer 1 (Friday). Substrate-coupled observer bandwidth bounded $\rightarrow$ correlations deviate from max-Tsirelson by $1/2^{N_c}$.

Section 8.5 — Falsifier Bell Experiment Design

SP-30 Substrate Engineering Program: Bell experiment design \$300-500K (Vienna/Caltech/Munich/Hanson). High-precision measurement of $|S|^2$ in CHSH inequality: - If $|S|^2 < 8$ by exactly $1/8$ (0.125) within experimental error $\rightarrow$ BST sub-Tsirelson confirmed - If $|S|^2$ saturates Tsirelson ($|S|^2 \rightarrow 8$ reliably) $\rightarrow$ BST refuted

This is one of BST's sharpest near-term operational falsifiers.

Section 8.6 — Calibration #17 RESOLVED

Thursday morning: Elie K52a Session 32 rank-1 projector framework ($B^2 = (126/16)|\psi_0\rangle\langle\psi_0|$) satisfies BOTH $\max=126/16 + Tr=126/16$; multi-month question refined to "WHICH substrate-natural $|\psi_0\rangle$ " pending Sessions 30+.

Section 8.7 — Honest scope

- Standard Bell-CHSH + Tsirelson recovered [\[?\]](#)
- BST sub-Tsirelson $1/8$ falsifier specified [\[?\]](#)
- Operational test at SP-30 Bell experiment design \$300-500K (pending Casey outreach decision)
- Calibration #17 operator-level closure multi-month (Elie Sessions 30+)

Section 8.8 — Connection to other chapters

- Vol 5 Ch 7 Born rule + measurement (K67 + T2479)
- Vol 5 Ch 11 POVMs + quantum information
- Vol 7 Information Theory (Bell-CHSH information-channel framing)
- T2399 + T2469 SCMP + Calibration #17

— Lyra, Vol 5 Ch 8 v0.3 chapter-grade narrative, Saturday 2026-05-23 morning EDT